

LAKEHEAD LMS PORTAL

CS5450 Mobile Programming - Challenge 2

Education / Learning Management System (LMS) Portal

Group: #1 - Education/LMS Portal

Instructor: Dr. Sabah Mohammed

Technology Stack: React Native, Expo, TypeScript, Firebase Authentication, Cloud Firestore

GitHub Repository:

<https://github.com/omavaiya4/LakeheadLMS>

1. Project Overview

Lakehead LMS Portal is a mobile Learning Management System (LMS) developed using React Native and Expo. The application provides a secure platform for students and instructors to manage academic activities including courses, grades, announcements, and user profiles.

The system uses Firebase Authentication for secure login and registration and Cloud Firestore for real-time data storage. The application supports role-based functionality, allowing students and instructors to access features relevant to their responsibilities.

2. Assignment Requirements Coverage

Requirement	Implementation
User Registration & Login	Firebase Authentication
Personalized Dashboard	Student and Instructor Dashboards
Search Functionality	Search Courses and Announcements
Responsive Design	Mobile Responsive React Native Interface
Security Integration	Authentication and Protected Access
Database Storage	Firebase Firestore
React Native / Expo	Fully Implemented
Android Support	Supported Through Expo

3. Key Features

Authentication

- User Registration
- User Login
- Secure Logout
- Session Persistence

Student Features

- View Courses
- View Grades
- Access Announcements
- Manage Profile

Instructor Features

- Create and Manage Courses
- Manage Student Grades
- Publish Announcements
- View Course Information

Search

- Search Courses
- Search Academic Information
- Quick Resource Navigation

4. Technology Stack

Frontend

- React Native
- Expo
- TypeScript
- Expo Router

Backend

- Firebase Authentication
- Cloud Firestore

Development Tools

- Visual Studio Code
- Expo CLI
- Android Studio Emulator
- GitHub

5. System Architecture

User

|



React Native Application

```

|
├── Authentication
├── Dashboard
├── Courses
├── Grades
├── Announcements
├── Search
└── Profile

```



Firestore Services

```

|
├── Authentication
└── Cloud Firestore

```

The mobile application communicates directly with Firebase services for authentication and data management.

6. Project Structure

LakeheadLMS

```

|
├── app
|   ├── (auth)
|   ├── (tabs)
|   ├── announcements
|   ├── course
|   └── profile
|
├── components
├── constants
├── hooks
├── assets
|
├── app.json
├── package.json
├── tsconfig.json
└── README.pdf

```

The project follows a modular structure that separates application screens, reusable components, configuration files, and business logic.

7. Security and Database

Firestore Authentication

The application uses Firebase Authentication to provide:

- Secure User Registration

- Secure User Login
- Session Management
- Role-Based Access

Cloud Firestore

The application stores data in Firestore collections:

users
courses
announcements
grades
enrollments

Firestore provides cloud-based storage and synchronization for all portal data.

8. Responsive Design

The user interface was designed specifically for mobile devices using React Native components and responsive layouts.

Responsive features include:

- Flexible Screen Layouts
- ScrollView Support
- Adaptive Components
- Consistent Navigation
- Android Compatibility

The application was tested using Android emulators and Expo.

9. Installation and Execution

Prerequisites

Install the following software:

- Node.js
- npm
- Expo CLI
- Android Studio

Clone Repository

```
git clone https://github.com/omavaiya4/LakeheadLMS.git  
cd LakeheadLMS
```

Install Dependencies

```
npm install
```

Configure Firebase

Open:

constants/firebase.ts

Add your Firebase project credentials.

Run Application

Start Expo:

`npx expo start`

Run Android Emulator:

a

Run Web Version:

w

10. Testing

The application was tested to verify functionality and stability.

Feature	Result
Registration	Passed
Login	Passed
Dashboard	Passed
Course Management	Passed
Grade Management	Passed
Search Functionality	Passed
Announcements	Passed
Firebase Integration	Passed

11. Challenges Encountered

During development, the following challenges were addressed:

- Firebase Authentication integration
- Firestore data management
- Role-based access implementation
- Responsive mobile interface design
- Navigation structure using Expo Router

These challenges were resolved through modular design and Firebase integration.

12. Future Improvements

Potential future enhancements include:

- Assignment Submission System
- Push Notifications
- Attendance Tracking
- File Upload Support
- Course Discussion Forums
- Dark Mode
- Real-Time Messaging

13. Conclusion

Lakehead LMS Portal successfully satisfies all requirements of the CS5450 Mobile Programming Challenge.

The application demonstrates the use of React Native, Expo, TypeScript, Firebase Authentication, and Cloud Firestore to build a secure and responsive Learning Management System. The implemented features include secure authentication, personalized dashboards, course management, grade tracking, announcements, search functionality, and cloud-based data storage.

Appendix - Screenshots

Figure 1 - Registration Screen

11:54



Create Account

Join LakeheadLMS today

I am a...



Student



Instructor

Full Name *



e.g. Om Avaiya

Email Address *



you@lakeheadu.ca

Password *



Min. 6 characters



Create Account

Already have an account? [Sign In](#)



11:54



Create Account

Join LakeheadLMS today

I am a...



Student



Instructor

Full Name *



e.g. Om Avaiya

Email Address *



you@lakeheadu.ca

Student ID



e.g. 1234567

Password *

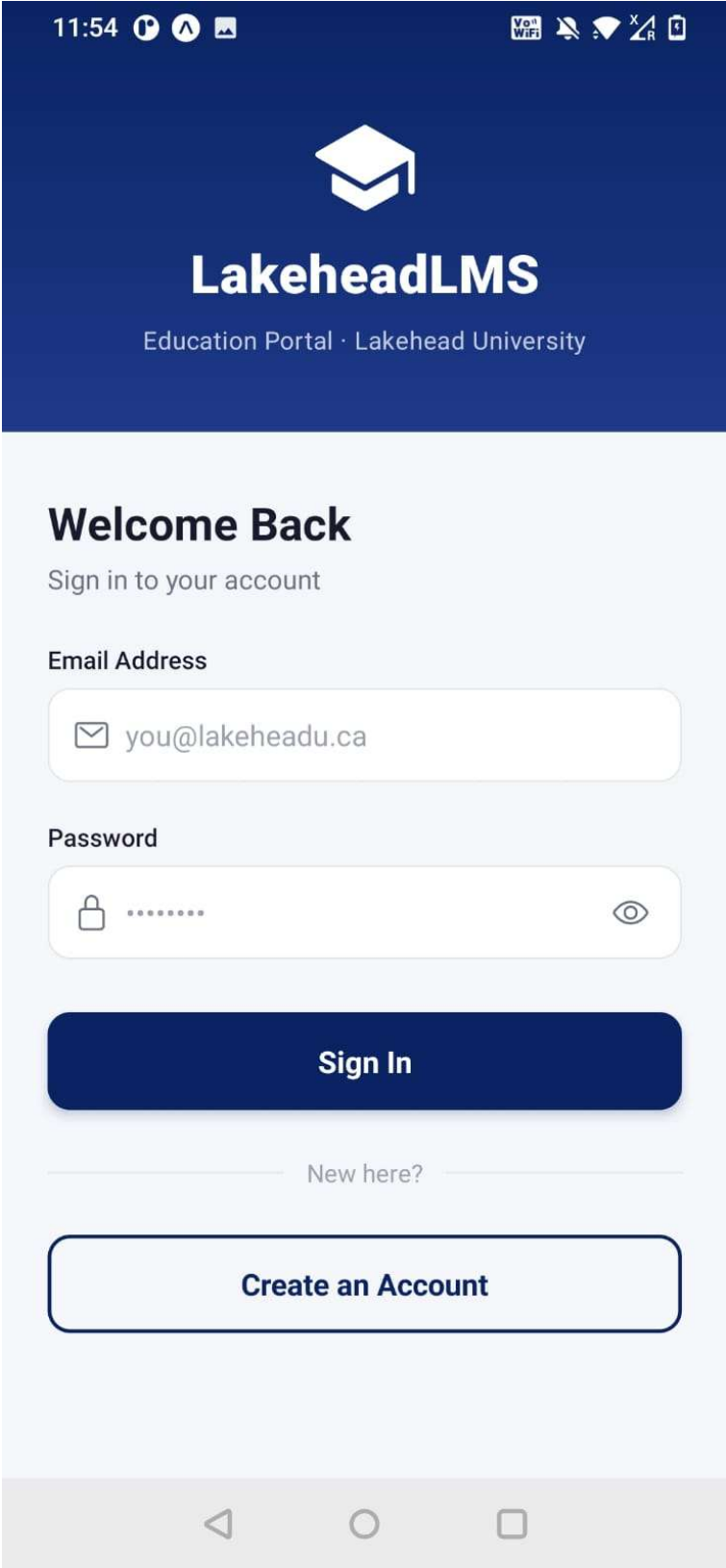


Min. 6 characters



Create Account

Figure 2 – Login Screen



The image shows a mobile application login screen for LakeheadLMS. The top section has a dark blue header with a white graduation cap icon, the text "LakeheadLMS", and "Education Portal · Lakehead University". Below this, the text "Welcome Back" is followed by "Sign in to your account". There are two input fields: "Email Address" with the placeholder "you@lakeheadu.ca" and "Password" with a masked password ".....". A dark blue "Sign In" button is below the password field. A link "New here?" is followed by a light blue "Create an Account" button. The bottom of the screen shows a grey bar with three navigation icons: a triangle, a circle, and a square.

11:54

Von Wi-Fi



LakeheadLMS

Education Portal · Lakehead University

Welcome Back

Sign in to your account

Email Address

 you@lakeheadu.ca

Password

Sign In

New here?

Create an Account

Figure 3 – Student Dashboard

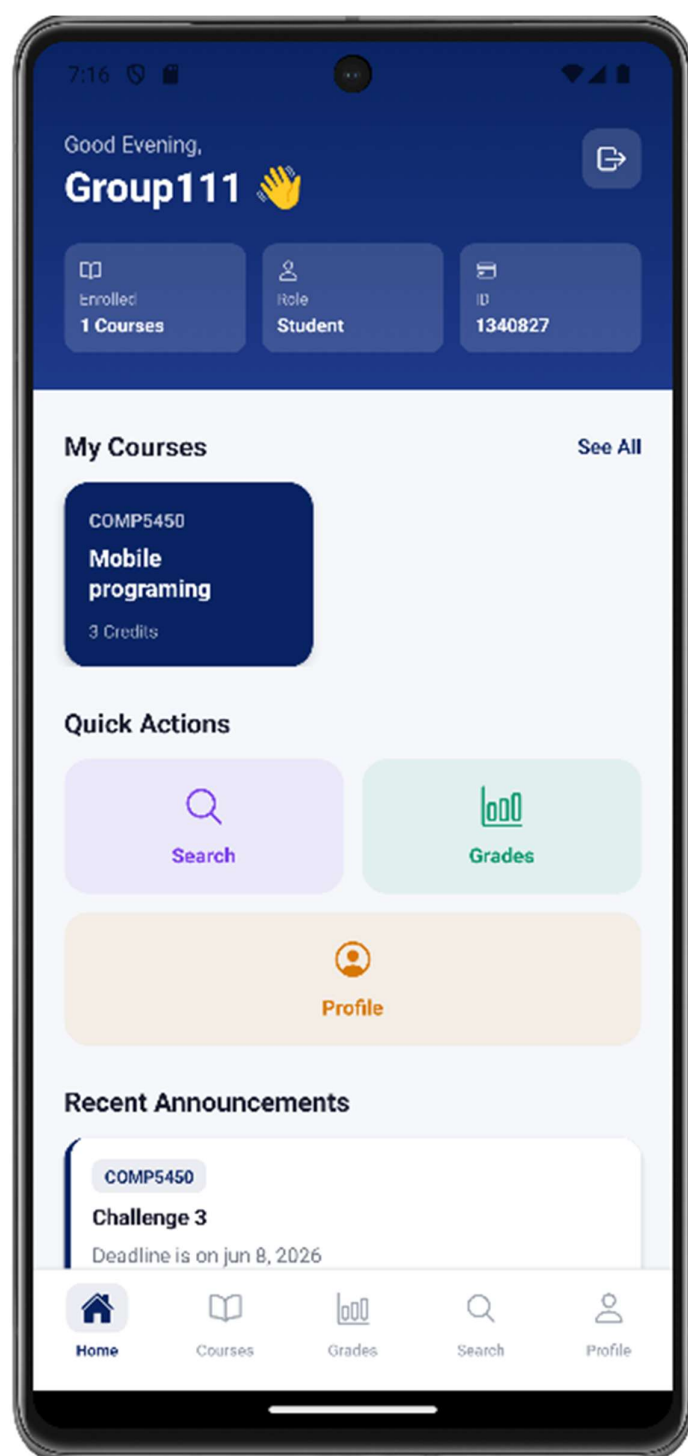


Figure 4 - Instructor Dashboard

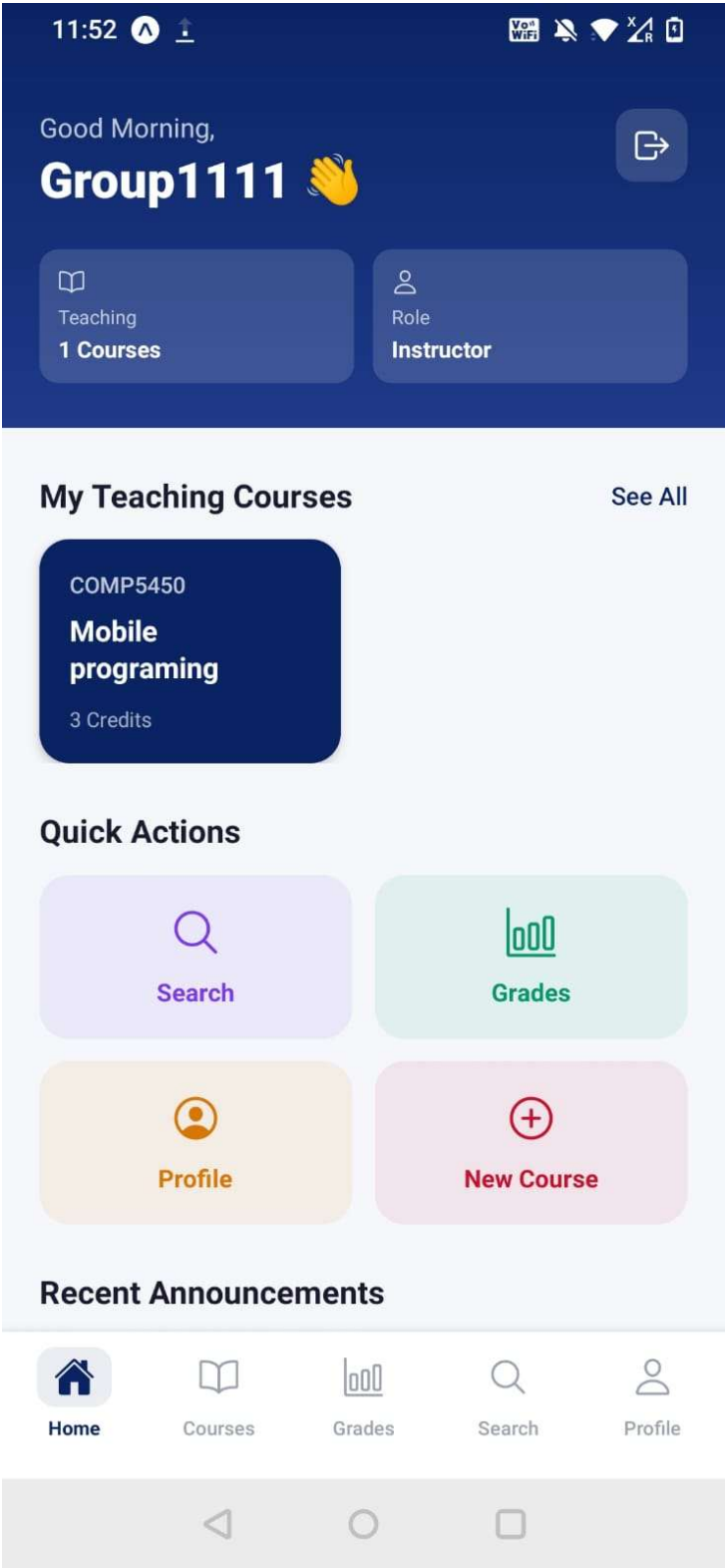


Figure 5 – Grade Management

Grade Management

+ Add Grade

COMP5450

Mobile Programming

ASSIGNMENT

Challenges 2

Group111

90/100

A+



Home



Courses



Grades



Search



Profile



Figure 6 – Search Functionality

Search

 Search courses, students, announcements...

All

Courses

Students

Announcements

Search

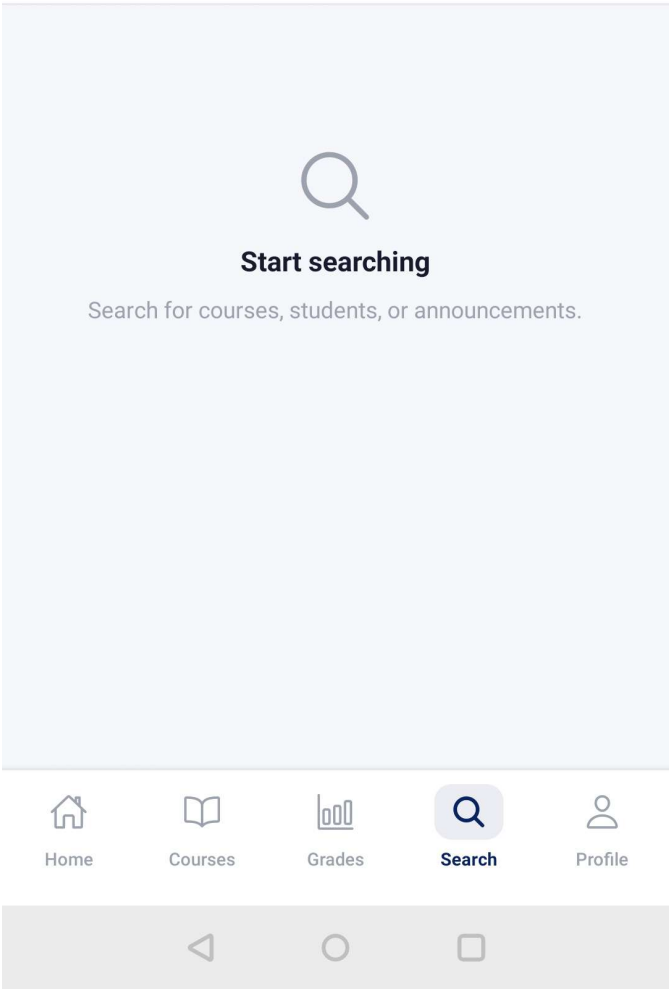


Figure 7 - Announcements

